

Date: 11/09/2026

Title: Mobile Number Validation using Exception Handling in Java

Objective / Aim

To design and implement a Java program to validate a mobile number using exception handling concepts. The program demonstrates the use of predefined exceptions such as `ArrayIndexOutOfBoundsException` and `NumberFormatException`, along with a user-defined exception called `LengthNotSufficientException`.

System Requirements

Hardware

- Standard PC/Laptop
- Intel Core i3/i5 processor or equivalent
- Minimum 4 GB RAM
- Minimum 500 MB free storage

Software

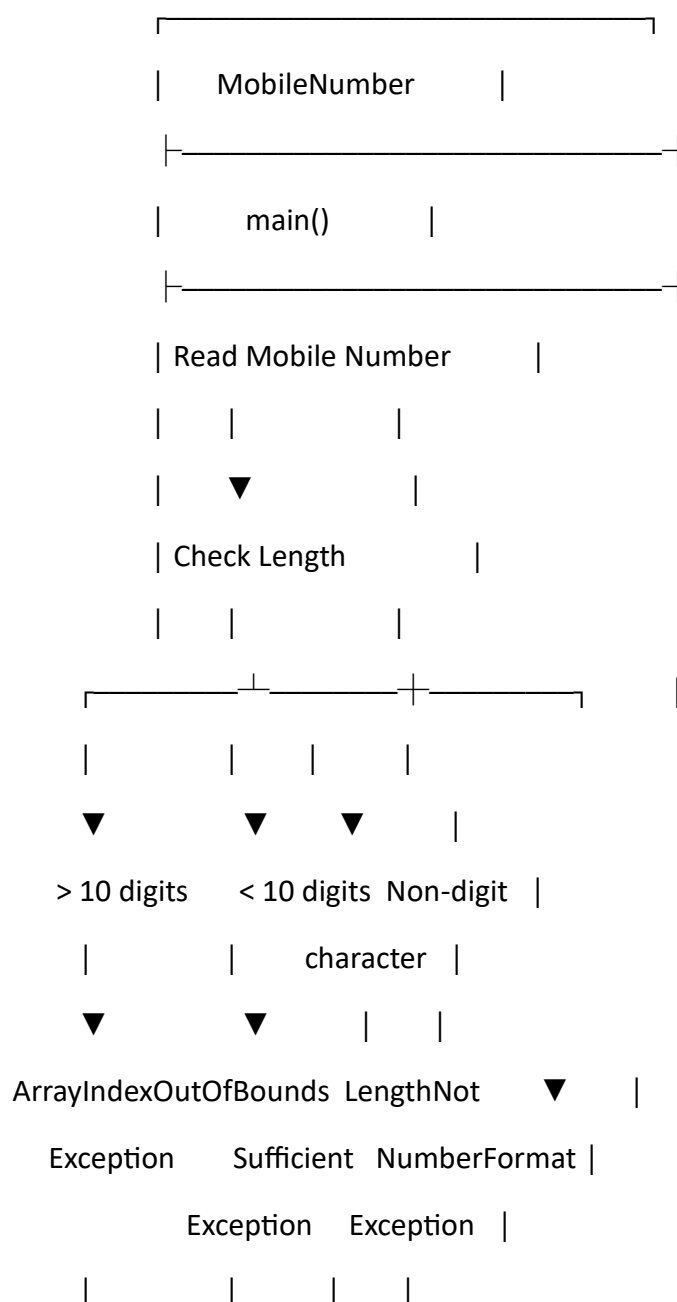
- Operating System: Windows / Linux / macOS
- Java Development Kit (JDK 17 or JDK 21)
- Text Editor or IDE such as Eclipse, IntelliJ IDEA, NetBeans, or VS Code
- Command Prompt / Terminal

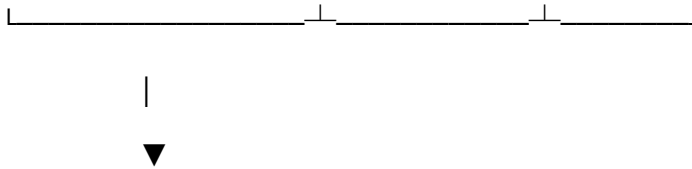
Algorithm / Logic Description

1. Create a package named `javapractice`.
2. Import the `Scanner` class from `java.util`.
3. Create a user-defined exception class named `LengthNotSufficientException`.
4. Extend the user-defined exception class from the `Exception` class.
5. Create a `MobileNumber` class containing the `main()` method.
6. Create a `Scanner` object to read the mobile number from the user.
7. Read the mobile number as a `String`.
8. Check whether the mobile number contains more than 10 digits.
9. If the length is greater than 10, throw `ArrayIndexOutOfBoundsException`.

10. Check whether the length is less than 10.
11. If the length is less than 10, throw `LengthNotSufficientException`.
12. Check each character of the mobile number using `Character.isDigit()`.
13. If any character is not a digit, throw `NumberFormatException`.
14. If all conditions are satisfied, display "Valid number".
15. Use separate catch blocks to handle each exception.
16. Display the appropriate error message based on the exception.

Exception Handling Structure





Exception Handling

Exception Types Used

- **ArrayIndexOutOfBoundsException** – Handles a mobile number containing more than 10 characters.
- **LengthNotSufficientException** – User-defined exception used when the mobile number contains fewer than 10 characters.
- **NumberFormatException** – Handles cases where the mobile number contains non-digit characters.



Source Code

```
package javapractice;
```

```
import java.util.Scanner;
```

```
// User-defined exception
```

```
class LengthNotSufficientException extends Exception {
```

```
    LengthNotSufficientException() {
```

```
        super("Invalid Mobile Number – LengthNotSufficientException");
```

```
    }
```

```
}
```

```
// Main class
```

```
public class MobileNumber {
```

```
    public static void main(String args[]) {
```

```
        // Create Scanner object
```

```
Scanner sc = new Scanner(System.in);

// Get mobile number from user
System.out.print("Enter Mobile Number: ");
String number = sc.next();

try {

    // Check for more than 10 digits
    if (number.length() > 10) {
        throw new ArrayIndexOutOfBoundsException();
    }

    // Check for less than 10 digits
    if (number.length() < 10) {
        throw new LengthNotSufficientException();
    }

    // Check each character
    for (int i = 0; i < 10; i++) {

        // If character is not a digit
        if (!Character.isDigit(number.charAt(i))) {
            throw new NumberFormatException();
        }
    }

    // Mobile number is valid
```

```

        System.out.println("Valid number");
    }

    // Catch excess length exception
    catch (ArrayIndexOutOfBoundsException e) {
        System.out.println(
            "Invalid Mobile Number-ArrayIndexOutOfBoundsException Exception"
        );
    }

    // Catch insufficient length exception
    catch (LengthNotSufficientException e) {
        System.out.println(e.getMessage());
    }

    // Catch non-digit exception
    catch (NumberFormatException e) {
        System.out.println(
            "Invalid Mobile Number – NumberFormatException"
        );
    }
}

```

Compilation & Execution Commands

Save the source code exactly as:

MobileNumber.java

Since the program uses:

```
package javapractice;
```

the file should be placed inside the javapractice folder.

Compile the program:

```
javac javapractice/MobileNumber.java
```

Run the compiled program:

```
java javapractice.MobileNumber
```

Output Screen

Case 1: Valid Mobile Number

Enter Mobile Number: 9876543210

Valid number

Case 2: Mobile Number with More Than 10 Digits

Enter Mobile Number: 98765432101

Invalid Mobile Number-ArrayIndexOutOfBoundsException

Case 3: Mobile Number with Less Than 10 Digits

Enter Mobile Number: 987654321

Invalid Mobile Number – LengthNotSufficientException

Case 4: Mobile Number Containing Non-Digit Characters

Enter Mobile Number: 98765abc10

Invalid Mobile Number – NumberFormatException

Result

The Mobile Number Validation program was successfully implemented in Java using exception handling. The program demonstrates the use of **predefined exceptions, user-defined exceptions, try-catch blocks, and character validation**. It successfully validates a mobile number and displays appropriate error messages for invalid length and non-digit characters.